

Performance Testing

Date	2 October 2026
Team ID	SWTID-2026-8368
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual timing with curl and server logs; pytest (24 tests) for functional checks
Type of Testing	End-to-end latency test (single user); no concurrent load because generation is GPU-bound
Target Module / API	POST /generate (outline, story, 5 images, PDF) and image_generator.generate_image
Test Environment	Local: Apple M5, 16 GB unified memory, PyTorch MPS, float16
Test Date	2 October 2026

Step 2: Test Scenarios

S. No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single panel image after warm-up (25 steps, 512 px)	1	About 12	One PNG saved
2	Full comic: submit form to preview (5 panels + PDF)	1	About 104	HTTP 200, 5-page PDF
3	Home page GET /	1	< 1	HTTP 200
4	Gemini quota exhausted	1	< 5	Clear HTTP 500 message

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	/ : < 0.1 s; /generate : 104 s	Pass (/ only)	Home page is fast; generation is AI-bound
2	Response Time (Max)	< 5 seconds	104 s (/generate)	Fail	Image generation dominates; 12.4 s per panel
3	Throughput (Req/sec)	-	About 0.01 comics/s (1 at a time)	Info	Single GPU worker
4	Error Rate	< 1%	0% in test runs; quota error handled	Pass	Gemini 429 shown clearly
5	CPU Utilization	< 80%	Not recorded (work runs on GPU/MPS)	Info	-
6	Memory Utilization	< 80%	Within 16 GB with fp16 and attention slicing	Pass	No out-of-memory

Step 4: Observations & Analysis

- Image generation is the bottleneck: about 12.4 s per panel on MPS after warm-up (5 panels = about 62 s).
- Story generation runs in parallel with images, so it adds no extra wait; the rest of the 104 s is Gemini latency, PDF and overhead.
- Start-up warm-up removes the one-time model load from the first request.
- Reducing steps or resolution was tested and reverted to keep image quality.

Step 5: Screenshots / Evidence

Preview page and generated panel from the end-to-end run:

Your Comic

Panel 1: The Threshold of Adventure



Preview page after the 104 s end-to-end run



A generated panel (Stable Diffusion v1.5)